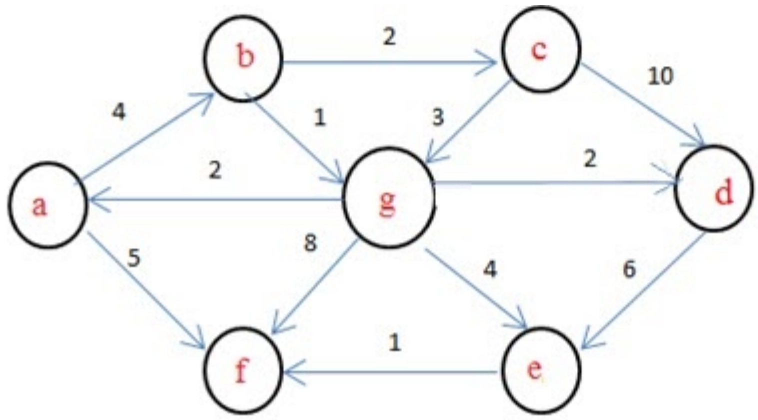
Data Structures and Algorithms Test

Data Structures and Algorithms Test

The respondent's email (**n0235147j@students.nust.ac.zw**) was recorded on submission of this form.

Consider the following graph.

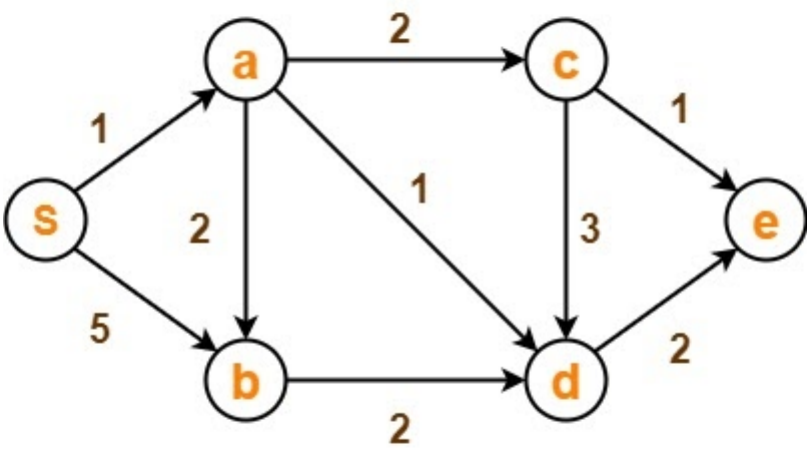
If b is the source vertex, what is the minimum cost to reach f vertex?



- ○ 9
- ◉ 6
- ○ 4
- ○ 8

Consider the gragh below

show the step by step process for obtaining the Single source shorterst path. Assume Node S as your source vertex



Using Dijkstra's Algorithm with S(S=0) as the source vertex, we would
1. Pick A from S, since A = 1 and B = 5
2. Picking A = 1, we have B = 3, C = 3, and D = 2
3. Picking D = 2, We get E being equal to 4
4. With B = 3 and C = 3 when at A, we find there is no change so our final shortest paths are

A = 1, B = 3, C = 3, D = 2, E = 4

What makes a tree 'binary'?

- (●) Each node can have at most two child nodes
- ( ) The three has two root nodes
- ( ) Each edge is bi directional
- ( ) Each leaf node has two parent nodes

In its most basic form, what does a node in a Linked List consist of?

- ○ Pointer to the head and tail of the linked list
- ○ An adjacency matrix
- ○ Data and Information
- ○ Arrows and pointers
- ◉ one area for data and one area with a pointer to the next node in the list

Which data structure allows us to access any element directly?

- ○ Priority queue
- ○ Queue
- ◉ Array
- ○ Stack
- ○ Linked List
- ○ Circular linked list

The worst case occurs in the Linear Search Algorithm when:

- ○ The item to be searched is in the last of the array
- ○ The item to be searched is not in the array
- ○ The item to be searched is in some where middle of the Array
- ◉ The item to be searched is either in the last or not in the array

The in-order and pre-order traversal of a binary tree are d b e a f c g and a b d e c f g respectively. The post order traversal of a binary tree is

- ○ e d b f g c a
- ○ e d b g f c a
- ○ d e f g b c a
- ◉ d e b f g c a

Which data structure do we use for testing a palindrome?

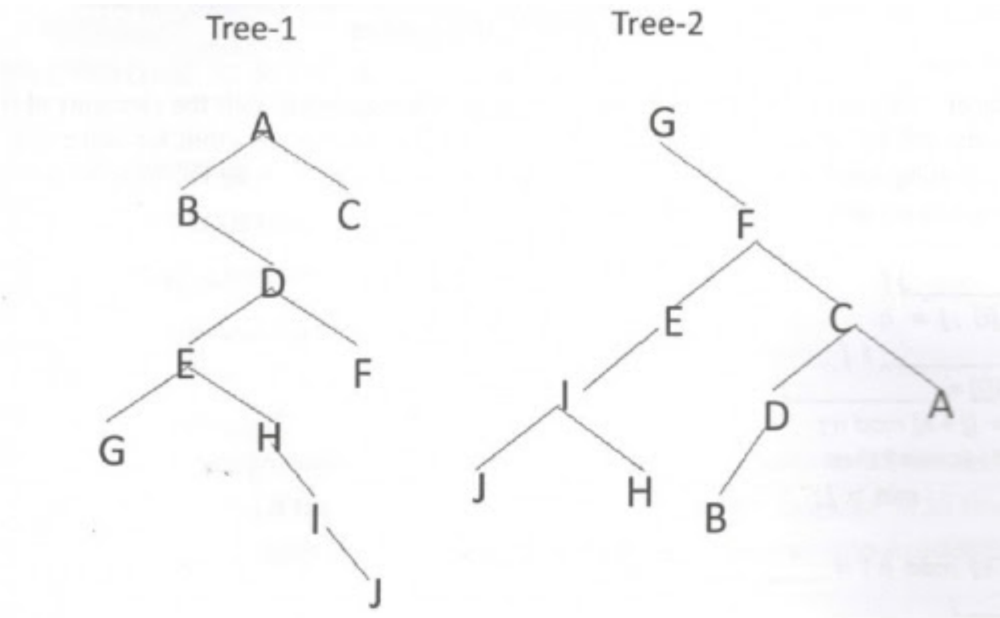
- ◉ Stack
- ○ Linked List
- ○ Priority queue
- ○ Tree
- ○ Heap

The most natural data structure for handling recursive function calls is

- ○ Circular Priority Queue
- ○ Linked List
- ○ Circular Queue
- ○ Array
- ○ Queue
- ◉ Stack

If Tree-1 and Tree-2 are the trees indicated below :

Which traversals of Tree-1 and Tree-2, respectively, will produce the same sequence?

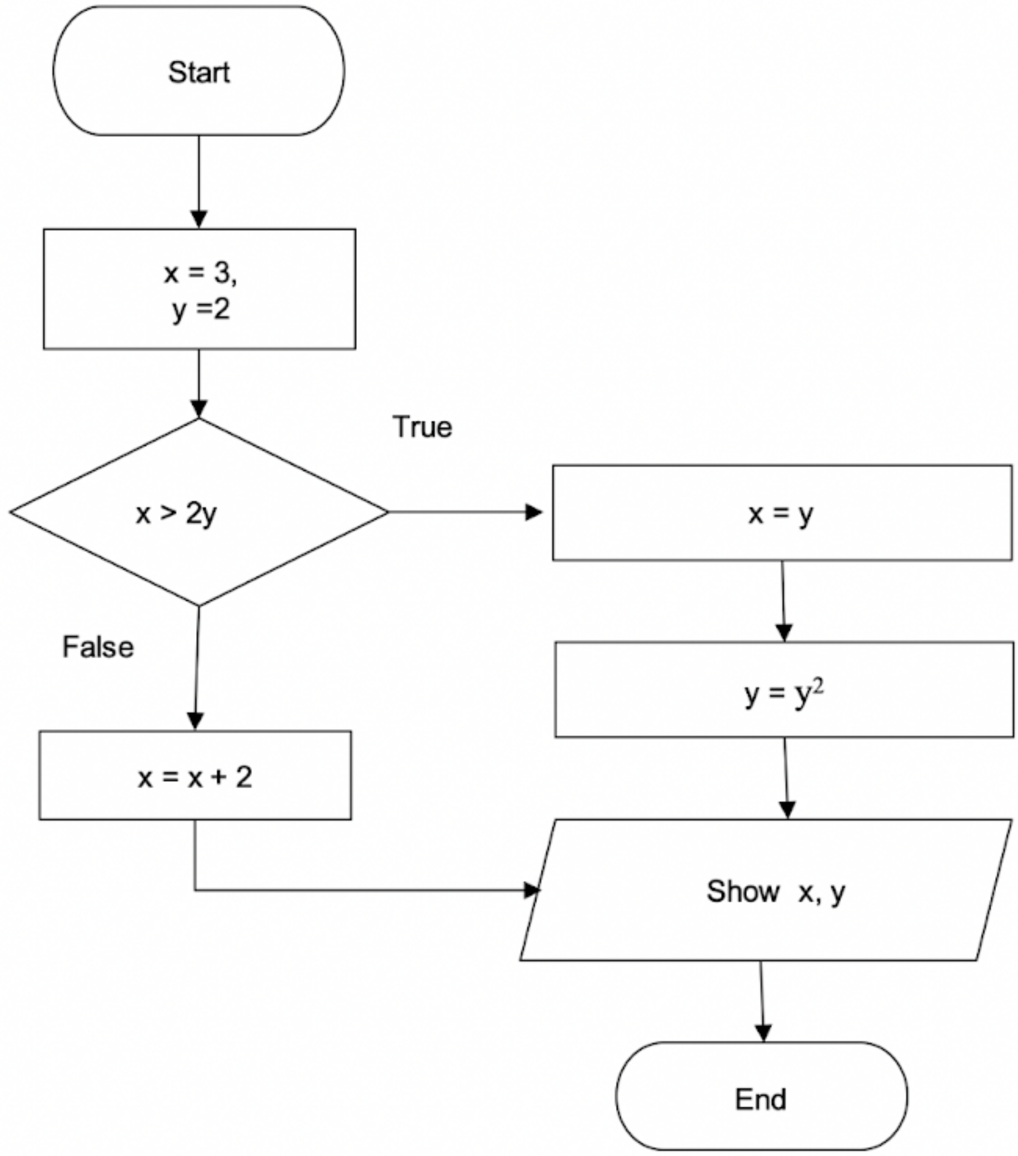


- [ ] Preorder, postorder
- [x] Postorder, preorder
- [ ] Inorder, preorder
- [ ] Postorder, inorder

Correct answer

- [x] Postorder, inorder

Convert the following flow-chart into an algorithm or a pseudo-code.



Start
Assign the integer 3 to variable x
Assign the integer 2 to variable y

```
IF x > 2 * y THEN
   x = y
   y = y * y
ELSE
   x = x + 2
END IF
```

Print x and y
End

All operations on a queue are

- [x] O(1)
- [ ] O(log n)
- [ ] O(n log n)
- [ ] O(n)

The average case occurs in the Linear Search Algorithm when:

- [x] The item to be searched is in some where middle of the Array
- [ ] The item to be searched is in the last of the array
- [ ] The item to be searched is either in the last or not in the array
- [ ] The item to be searched is not in the array

What kind of graph does Dijkstra's Algorithm not work with?

- [x] Graphs with negative weights
- [ ] Weighted
- [ ] Unidirected
- [ ] Directed

The inorder and preorder Traversal of binary Tree are dbeafcg and abdecfg respectively. The post-order Traversal is __________.

- (selected) debfgca
- dbefacg
- dbefcga
- debfagc

Choose the equivalent prefix form of the following expression (a + (b − c))* ((d − e)/(f + g − h))

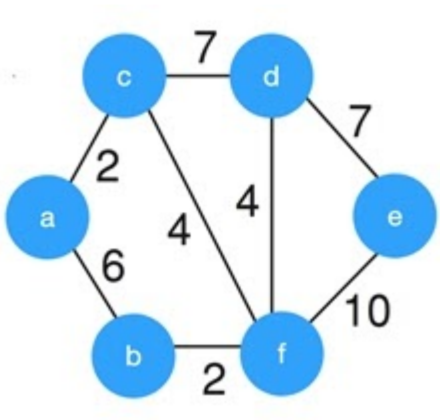
- (selected) * +a −bc − /de − +fgh
- * +a − bc /− ed + −fgh
- * +a − bc /− de − +fgh
- * +ab − c /− ed + −fgh

Correct answer

- (selected) * +a − bc /− de − +fgh

Consider the given graph:

What is the weight of the minimum spanning tree using the Kruskal's algorithm?



- ( ) 15
- ( ) 24
- (•) 19
- ( ) 23

No correct answers

Big-O notation expresses

- ( ) Average Case
- ( ) Lower bounds
- ( ) Best case
- (•) Upper bounds

Which of the following methods can be used to solve the Knapsack problem?

- ( ) Brute force algorithm
- ( ) Dynamic programming
- ( ) Recursion
- (•) Brute force, Recursion and Dynamic Programming

Which of these is a divide and conquer algorithm?

- [x] Merge Sort
- [ ] Bubble Sort
- [x] Quick Sort
- [ ] Selection Sort

If a binary search tree is not --------------------- it may be less efficient than a linear structure.

- ( ) complete
- ( ) Hashed
- ( ) empty
- ( ) None of the above
- (•) balanced

What is the worst case time complexity of a linear search?

- (x) O(n)
- ( ) O(n log n)
- ( ) O(n^2)
- ( ) O(log n)

Replacing a value at the kth position of an array (A[k]) is

- ( ) O(log n)
- ( ) O(n)
- (x) O(1)
- ( ) O(n 2)

The best technique for handling collision is:

- ( ) Double probing
- ( ) Double hashing
- ( ) Linear probing
- (x) Separate chaining
- ( ) Quadratic probing

Correct answer

- (x) Quadratic probing

In a Max-Heap, each node's priority value

- ◯ Smaller or larger than all its children
- ◯ None of the above.
- ◉ Larger than all its children
- ◯ Smaller than all its children

Given items as {value,weight} pairs {{40,20},{30,10},{20,5}}. The capacity of knapsack=20. Find the maximum value output assuming items to be divisible.

- ◯ 60
- ◉ 80
- ◯ 40
- ◯ 100

Correct answer

- ◉ 60

What is the best sorting algorithm to use for the elements in array are more than 1 million in general?

- ( ) Heap Sort
- ( ) Insertion sort.
- ( ) Bubble sort.
- (•) Merge sort.
- ( ) Quick sort.

Correct answer

- (•) Quick sort.

Dijkstra's Algorithm is used to solve ___________ problems.

- ( ) All pair shortest path
- ( ) Network flow
- ( ) Sorting
- (•) Single source shortest path

Given items as {value,weight} pairs {{60,20},{50,25},{20,5}}. The capacity of knapsack=40. Find the maximum value output assuming items to be divisible

- ( ) 130,
- ( ) 110, 70
- (•) 110
- ( ) 100

The 0-1 Knapsack problem can be solved using Greedy algorithm.

- ( ) True
- (•) False

Linear search is also called------

- [x] Sequential search
- [ ] None
- [ ] Perfect search
- [ ] Random Search

Dijkstra’s Algorithm is the prime example for ___________

- (selected) Greedy algorithm
- Branch and bound
- Back tracking
- Dynamic programming

You are given a knapsack that can carry a maximum weight of 60. There are 4 items with weights {20, 30, 40, 70} and values {70, 80, 90, 200}. What is the maximum value of the items you can carry using the 0/1 knapsack?

- 160
- (selected) 170
- 90
- 200

Correct answer

- (selected) 160

The in-order traversal of a tree resulted in FBGADCE. Then the pre-order traversal of that tree would result in

- ABFGCDE
- FGBDECA
- BFGCDEA
- (selected) AFGBDEC

Correct answer

- (selected) ABFGCDE

Kruskal's algorithm is best suited for the dense graphs than the prim's algorithm.

- (selected) False
- True

Suppose we are sorting an array of eight integers using quicksort, and we have just finished the first partitioning with the array looking like this:
3 0 2 5 1 7 9 12 11 100 10

Which statement is correct?

- ( ) Neither the 7 nor the 9 is the pivot.
- ( ) The pivot is not the 7, but it could be the 9
- ( ) The pivot could be either the 7 or the 9.
- (•) The pivot could be the 7, but it is not the 9

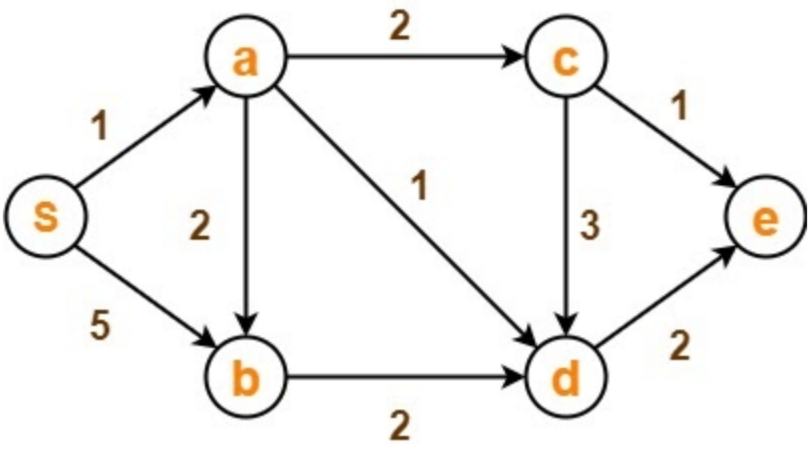
Correct answer

- (•) The pivot could be either the 7 or the 9.

How is a circular Linked List created?

- ( ) By rotating the head and tail nodes
- (•) By pointing to the head node from the tail node's 'next' pointer
- ( ) By flipping each node except the last one
- ( ) By inserting a special rotation data element

Consider the graph below. Use Kruskal's algorithm to determine the MST. show the step by step process



Firstly, we sort edges in ascending order, Getting
SA = 1, AD = 1, CE = 1, AC = 2, AB = 2, BD = 2, DE = 2, CD = 3, SB = 5

Next we then pick edges without forming a circle, getting
SA = 1, AD = 1, CE = 1, AC = 2, AB = 2

We stop when all vertices are now connected by n-1 edges
So our MST = SA, AD, CE, AC, AB(With total weight 7)

What algorithm is used to find the shortest path in a graph?

- [ ] Kruskal's Algorithm
- [ ] Depth First Search
- [ ] Prim's Algorithm
- [x] Dijkstra's algorithm
- [x] Bellman–Ford algorithm

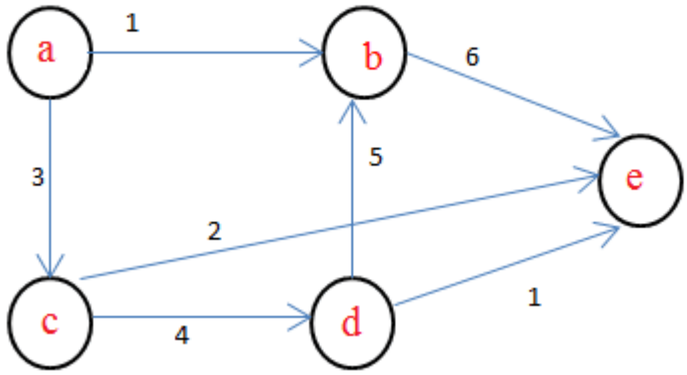
The post-order traversal of a binary search tree is given by 2, 7, 6, 10, 9, 8, 15, 17, 20, 19, 16, 12. Then the pre-order traversal of this tree is:

- ◯ 7, 6, 2, 10, 9, 8, 15, 16, 17, 20, 19, 12
- ◯ 2, 6, 7, 8, 9, 10, 12, 15, 16, 17, 19, 20
- ◯ 7, 2, 6, 8, 9, 10, 20, 17, 19, 15, 16, 12
- ◉ 12, 8, 6, 2, 7, 9, 10, 16, 15, 19, 17, 20

Fractional knapsack problem is solved most efficiently by which of the following algorithm?

- ◉ Greedy algorithm
- ◯ Backtracking
- ◯ Dynamic programming
- ◯ Divide and conquer

In the given graph, identify the shortest path having minimum cost to reach vertex E if A is the source vertex.



- [x] a-c-e
- [ ] a-b-e
- [ ] a-c-d-b-e
- [ ] a-c-d-e

Which of following option is not correct regarding depth first searching?

- [ ] Depth-first search requires O(V^2) time if implemented with an adjacency matrix.
- [ ] None of these
- [ ] Depth-first search requires O(V + E) time if implemented with adjacency lists
- [x] In a depth-first traversal of a graph G with V vertices, E edges are marked as tree edges. The number of connected components in G is (E - V).

The necessary condition for using binary search in an array is :-

- (●) The array should be sorted
- ( ) The array should be of a big size
- ( ) The array should not be too big
- ( ) The array should be sorted in ascending order

Which of the following is true?

- (●) Kruskal's algorithm can also run on the disconnected graphs
- ( ) Prim's algorithm can also be used for disconnected graphs
- ( ) Prim's algorithm is simpler than Kruskal's algorithm
- ( ) In Kruskal's sort edges are added to MST in decreasing order of their weights

Postorder traversal of a binary tree will visit

- ( ) Node, then its children
- (●) Node's children, then the node
- ( ) No particular order
- ( ) Node, left child, then right child

Correct answer

- (●) Node, left child, then right child

A solution to a problem is said to be efficient if

- (●) the problem is solved within the required resource constraints.
- ( ) problem is solved in constant (O(1)) time.
- ( ) does not solve the problem.
- ( ) problem is solved regardless of the needed resources.

What kind of graph does Bellman-Ford's Algorithm work with?

- [x] Graphs with negative weights
- [x] Directed
- [x] Weighted
- [ ] Unidirected

Correct answer

- [x] Weighted
- [x] Directed
- [x] Unidirected
- [x] Graphs with negative weights

Which of the following sorting algorithms has the lowest worst-case complexity?

- ( ) Bubble Sort
- ( ) Selection Sort
- ( ) Quick Sort
- (•) Merge Sort
- ( ) Interpolation Sort

What data structure uses LIFO?

- ( ) Queue
- ( ) Array
- (•) Heap Data Structure
- ( ) Priority Queue
- ( ) Linked List

Correct answer

- (•) Queue

What is the objective of the knapsack problem?

- ( ) To get minimum total value in the knapsack
- ( ) To get maximum weight in the knapsack
- (•) To get maximum total value in the knapsack
- ( ) To get minimum weight in the knapsack

Do not edit this field, click on SUBMIT

The field below is needed by Form Timer and must NOT be modified. If you modify this key your answer will not be recorded.

Fingerprint - DO NOT EDIT *